

Pearson Correlation for Ordinal Variables: Consolidated Analysis

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1 Extremal Pearson Correlation for Ordinal Variables

1.1 1. Problem Setting and Notation

We consider two ordinal variables:

- X with K_X ordered categories $0, 1, \dots, K_X - 1$
- Y with K_Y ordered categories $0, 1, \dots, K_Y - 1$

Marginals given by:

$$p_X(i) = P(X = i), \quad p_Y(j) = P(Y = j)$$

or counts:

$$n_{X=i}, \quad n_{Y=j}, \quad N = \sum_i n_{X=i} = \sum_j n_{Y=j}$$

CDFs:

$$F_X(i) = \sum_{k \leq i} p_X(k), \quad F_Y(j) = \sum_{k \leq j} p_Y(k)$$

Pearson correlation:

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad E[XY] = \sum_{i=0}^{K_X-1} \sum_{j=0}^{K_Y-1} x_i y_j P(X = x_i, Y = y_j)$$

For Spearman's correlation, replace X, Y by their ranks R_X, R_Y .

(Sources: *Proofs-OrdinalVariables.md*; *Ordinal Variables and Correlation.md*; *BFbounds.md*; *Math Proof GPT – Whitt 1976 Paper.md*)

1.2 2. Fréchet–Hoeffding / Boole–Fréchet Bounds

Any joint CDF F_{XY} with marginals F_X, F_Y satisfies:

$$\max\{F_X(i) + F_Y(j) - 1, 0\} \leq F_{XY}(i, j) \leq \min\{F_X(i), F_Y(j)\}$$

(*BFbounds.md*)

- **Upper bound (comonotonic):**

$$F_{XY}^{\text{upper}}(i, j) = \min\{F_X(i), F_Y(j)\}$$

- **Lower bound (countermonotonic):**

$$F_{XY}^{\text{lower}}(i, j) = \max\{F_X(i) + F_Y(j) - 1, 0\}$$

1.3 3. Maximizing and Minimizing r

1.3.0.1 3.1 Max r — Comonotonic Coupling

Claim: The maximum correlation $r_{\max}$ is achieved by the comonotonic (FH upper) arrangement.

Proofs / Arguments: - **FH upper pmf construction:** In discrete case, differences of CDF steps allocate mass along the diagonal, pairing high X with high Y .

(*BFbounds.md*; *Proofs-OrdinalVariables.md*) - **Rearrangement inequality (ranks):** For ranks R_X, R_Y , $\sum R_X R_Y$ is maximized when sequences are ordered the same way.

(*Ordinal Variables and Correlation.md*) - **Quantile coupling:** $U \sim \text{Unif}(0, 1)$, $X = F_X^{-1}(U)$, $Y = F_Y^{-1}(U)$.

(*Math Proof GPT – Whitt 1976 Paper.md*)

1.3.0.2 3.2 Min r — Countermonotonic Coupling

Claim: The minimal correlation $r_{\min}$ corresponds conceptually to the countermonotonic (FH lower) arrangement.

Proofs / Arguments: - **FH lower pmf construction:** Differences of the lower-bound CDF.

(*BFbounds.md*) - **Opposite-order pairing:** Sort X descending, Y ascending; achieves best feasible negative correlation.

(*Proofs-OrdinalVariables.md*; *Ordinal Variables and Correlation.md*) - **Quantile coupling:** $X = F_X^{-1}(U)$, $Y = F_Y^{-1}(1 - U)$ in ideal continuous case.

(*Math Proof GPT – Whitt 1976 Paper.md*)

Note: In discrete settings, FH lower may not be attainable; see Section 5.

1.4 4. Binary (2×2) Cases and Examples

(*From Cheemera v1.1 – Pearson Correlation Analysis Implications.md*)

1.4.0.1 4.1 Balanced marginals (70–30, 70–30)

- Perfect diagonal fill yields $r_{\max} = 1$.

1.4.0.2 4.2 Skew vs balanced (90–10, 50–50)

- Perfect alignment impossible; $r_{\max} < 1$ due to capacity limits.

1.5 5. Attainability and Symmetry (Updated)

1.5.1 5.1 Maximal correlation

- **Result:** In the discrete ordinal setting, $r_{\max}$ is **always** attained by the FH upper (comonotonic) construction.

(*BFbounds.md*; *Proofs-OrdinalVariables.md*; *Ordinal Variables and Correlation.md*)

1.5.2 5.2 Minimal correlation

- **Result:** The FH lower (countermonotonic) construction may be **unattainable** in discrete settings; $r_{\min}$ is then the best feasible opposite-order arrangement.
(*BFbounds.md; Proofs-OrdinalVariables.md; Ordinal Variables and Correlation.md*)

1.5.3 5.3 Magnitude relationships

- **Earlier view:** Often $|r_{\min}| < r_{\max}$ for discrete asymmetrical marginals.
(*BFbounds.md; Ordinal Variables and Correlation.md*)
- **New evidence (BES data):** Possible to have $|r_{\min}| > r_{\max}|$.
Example: X with 4 categories, Y with 5 categories, skewed marginals;
 $r_{\max} = 0.85027$, $r_{\min} = -0.89916$.
(*Cees - r_min MAGNITUDE Asymmetry in Pearson Correlation.md*)

Reason: Pearson correlation uses centered values; skewness can make negative deviations in antitone pairing larger in magnitude than positive deviations in comonotone pairing.

1.5.4 Lemma (signs of extreme correlations under monotone scoring)

Let X and Y be ordinal variables. Assign numeric scores to categories via **non-decreasing** maps s_X, s_Y that respect the category order (scores may be affine or nonlinear; ties allowed). Define two extreme couplings:

- **Comonotone (upper):** pair larger scored values of X with larger scored values of Y .
- **Antitone (lower):** pair larger scored values of X with smaller scored values of Y .

Let $r_{\max}$ and $r_{\min}$ be the Pearson correlations computed from these couplings, either (i) in the **finite-sample/permutation** sense (sort scored observations in the same vs. opposite order), or (ii) in the **Fréchet–Hoeffding (FH)/population** sense (pair quantiles $Q_X(u)$ with $Q_Y(u)$ vs. $Q_Y(1 - u)$).

Then:

1. $r_{\max} \geq 0$ and $r_{\min} \leq 0$.
2. Moreover, $r_{\max} > 0$ and $r_{\min} < 0$ **iff** both variables have **positive variance** under the chosen scoring (i.e., neither scored variable is almost surely constant).

(If you reverse the scoring of one variable (decreasing map), swap “upper” and “lower”. For Spearman’s

$$\rho$$

, take the scores to be midranks—monotone by construction—and the same conclusions hold.)

1.6 Proof

We give parallel proofs in the finite-sample and population settings.

1.6.1 A) Finite-sample (permutation) setting

Let $x_{(1)} \geq \dots \geq x_{(n)}$ and $y_{(1)} \geq \dots \geq y_{(n)}$ be the **scored** observations of X and Y sorted in nonincreasing order (monotone scoring ensures sorting by category equals sorting by score). Center them:

$$x'_i := x_{(i)} - \bar{x}, \quad y'_i := y_{(i)} - \bar{y}, \quad \bar{x} := \frac{1}{n} \sum_i x_{(i)}, \quad \bar{y} := \frac{1}{n} \sum_i y_{(i)}.$$

Define the two extreme centered sums

$$S_{\max} := \sum_{i=1}^n x'_i y'_i, \quad S_{\min} := \sum_{i=1}^n x'_i y'_{n+1-i}.$$

- **Signs.**

Chebyshev's sum inequality (rearrangement for similarly/ oppositely ordered sequences) gives

$$\frac{1}{n} \sum_i x'_i y'_i \geq \left(\frac{1}{n} \sum_i x'_i \right) \left(\frac{1}{n} \sum_i y'_i \right) = 0, \quad \Rightarrow S_{\max} \geq 0,$$

and, because y'_{n+1-i} is **nondecreasing** in i ,

$$\frac{1}{n} \sum_i x'_i y'_{n+1-i} \leq \left(\frac{1}{n} \sum_i x'_i \right) \left(\frac{1}{n} \sum_i y'_{n+1-i} \right) = 0, \quad \Rightarrow S_{\min} \leq 0.$$

Dividing by $\sigma_X \sigma_Y > 0$ (the standard deviations of the scored variables) yields $r_{\max} \geq 0$ and $r_{\min} \leq 0$.

- **Strictness ($\Leftrightarrow$ positive variances).**

Suppose $\sigma_X > 0$ and $\sigma_Y > 0$ (i.e., neither $\{x'_i\}$ nor $\{y'_i\}$ is identically zero). The rearrangement inequality says $S_{\max}$ is the **maximum** and $S_{\min}$ the **minimum** of $\sum_i x'_i y'_{\pi(i)}$ over all permutations π ; moreover $S_{\min} < S_{\max}$. The **average** over all permutations equals

$$\frac{1}{n!} \sum_{\pi} \sum_i x'_i y'_{\pi(i)} = \sum_i x'_i \left(\frac{1}{n!} \sum_{\pi} y'_{\pi(i)} \right) = \sum_i x'_i \cdot \frac{1}{n} \sum_j y'_j = 0.$$

Since the minimum is $<$ the maximum and the average is 0, it follows that $S_{\min} < 0 < S_{\max}$. Hence $r_{\min} < 0 < r_{\max}$.

Conversely, if $\sigma_X = 0$ or $\sigma_Y = 0$, then one centered sequence is identically zero, so $S_{\max} = S_{\min} = 0$ and the correlations are undefined or (in a limiting sense) attain 0. This matches the “only if” direction.

1.6.2 B) Population (FH/quantile) setting

Let $Q_X, Q_Y : [0, 1] \rightarrow \mathbb{R}$ be the quantile functions of the **scored** variables. Set

$$f(u) := Q_X(u) - \mu_X, \quad g(u) := Q_Y(u) - \mu_Y,$$

so $\int_0^1 f = \int_0^1 g = 0$. Monotone scoring preserves category order, hence Q_X, Q_Y are nondecreasing; therefore f, g are nondecreasing functions.

- **Upper/comonotone covariance:**

$$\text{Cov}_{\max} = \int_0^1 f(u) g(u) du \geq \left(\int_0^1 f \right) \left(\int_0^1 g \right) = 0$$

by Chebyshev’s **integral** inequality for similarly ordered functions.

- **Lower/antitone covariance:**

$$\text{Cov}_{\min} = \int_0^1 f(u) g(1-u) du \leq \left(\int_0^1 f \right) \left(\int_0^1 g(1-u) \right) = 0,$$

since f is nondecreasing and $g(1-u)$ is **nonincreasing**.

Dividing by $\sigma_X \sigma_Y > 0$ yields $r_{\max} \geq 0$ and $r_{\min} \leq 0$.

- **Strictness.**

If both variables have positive variance, then f and g are not a.e. constant. For nontrivial nondecreasing f, g , Chebyshev’s integral inequalities above are **strict** unless one function is a.e. constant. Hence $\text{Cov}_{\max} > 0$ and $\text{Cov}_{\min} < 0$, giving $r_{\max} > 0$ and $r_{\min} < 0$. Conversely, if either variance is 0, both covariances are 0.

This completes the proof. $\square$

1.6.3 remarks (orientation & invariance)

- The lemma depends only on **order**, not spacing. Any (non-decreasing) rescaling—including affine changes like 10, 20, 30, ... or nonlinear monotone spacings—preserves the sign conclusions.
- If you encode one variable with a **decreasing** scoring map, then “comonotone” in category order becomes **antitone** in numeric order; swap the labels to recover the same sign statements.
- For **Spearman**, take scores as midranks; everything above applies verbatim.

1.7 5.4 Symmetry condition

- If one marginal is **palindromically symmetric** about its center rank, then $r_{\min} = -r_{\max}$.
(*Cees - r_min MAGNITUDE Asymmetry in Pearson Correlation.md*)

1.8 necessary/sufficient symmetry condition (precisely stated)

- **Pair-specific equality** $|r_{\min}| = r_{\max} \iff \sum_i x'_i (y'_i + y'_{n+1-i}) = 0$.
(Can happen by coincidence even if Y isn't symmetric.)
- **Universal in X equality** (for **all** X):
Necessary and sufficient that **one** marginal (say Y) has
 - (i) **palindromic mass symmetry** $p_Y(k) = p_Y(K_Y - 1 - k)$ and
 - (ii) **antisymmetric scoring** $s_Y(k) + s_Y(K_Y - 1 - k) = \text{const}$ (true for any affine map of $0:(K_Y - 1)$).
This condition is **independent of K_X vs K_Y** .

1.9 Symmetry

1.9.1 1) setup (permutation extremes for a single sample)

Let $x_{(1)} \geq \dots \geq x_{(n)}$ and $y_{(1)} \geq \dots \geq y_{(n)}$ be the sorted sample values (or expanded from counts). Define centered sequences

$$x'_i := x_{(i)} - \bar{x}, \quad y'_i := y_{(i)} - \bar{y},$$

and the two permutation extremes

$$S_{\max} = \sum_{i=1}^n x'_i y'_i, \quad S_{\min} = \sum_{i=1}^n x'_i y'_{n+1-i},$$

with $r_{\max} = S_{\max}/(\sigma_X \sigma_Y)$, $r_{\min} = S_{\min}/(\sigma_X \sigma_Y)$.

We always have $S_{\max} \geq 0$ and $S_{\min} \leq 0$. Magnitude symmetry $|r_{\min}| = r_{\max}$ is equivalent to

$$S_{\max} + S_{\min} = 0.$$

1.9.2 2) pair-specific necessary & sufficient condition

Let

$$z_i := y'_i + y'_{n+1-i}.$$

Then

$$S_{\max} + S_{\min} = \sum_{i=1}^n x'_i z_i.$$

Hence, for this *particular* pair (x, y) ,

$$\boxed{|r_{\min}| = r_{\max} \iff \sum_i x'_i z_i = 0} \quad ()$$

(i.e., the centered X sequence is orthogonal to the “palindrome-sum” vector z).

This can hold even if Y is not symmetric—by coincidence of that inner product being zero.

1.9.3 3) universal condition (independent of X)

If you want $|r_{\min}| = r_{\max}$ to hold for **every** possible X (every centered, sorted x'), then () must hold for all x' . The only way is

$$z_i \equiv 0 \quad \text{for all } i,$$

i.e.

$$y'_i + y'_{n+1-i} \equiv 0 \quad \forall i.$$

Translating back to categories and scores, a **necessary and sufficient universal condition** is:

- **Palindromic mass symmetry for Y :** $p_Y(k) = p_Y(K_Y - 1 - k)$ for all k , and
- **Antisymmetric scoring for Y :** the score map satisfies $s_Y(k) + s_Y(K_Y - 1 - k)$ is constant (true for affine scores like $0, 1, \dots, K_Y - 1$, and for Spearman midranks under palindromic counts).

Under these, $z_i \equiv 0$ and thus $|r_{\min}| = r_{\max}$ for any X .

If either mass symmetry or score antisymmetry fails, you can generally find an X for which $|r_{\min}| \neq r_{\max}$ (often $|r_{\min}| > r_{\max}$ when Y is skewed).

1.9.4 4) does the number of categories matter?

- **For equality itself:** No. Odd vs even, and equal vs unequal numbers of categories, do **not** affect whether $|r_{\min}| = r_{\max}$ can hold. What matters is the symmetry/antisymmetry described above.
- **For the sizes of the extremes:** Different K , skewness, and spacing of scores *do* affect how large $r_{\max}$ and $|r_{\min}|$ can be, but not whether the magnitudes are equal.

1.9.5 5) FH (transport) version

For FH extremes (optimize over all joints with given marginals), replace sums by integrals over quantiles:

$$\text{Cov}_{\max} = \int_0^1 (Q_X(u) - \mu_X)(Q_Y(u) - \mu_Y) du, \quad \text{Cov}_{\min} = \int_0^1 (Q_X(u) - \mu_X)(Q_Y(1-u) - \mu_Y) du.$$

Then

$$\text{Cov}_{\max} + \text{Cov}_{\min} = \int_0^1 (Q_X(u) - \mu_X) [Q_Y(u) + Q_Y(1-u) - 2\mu_Y] du.$$

So the **pair-specific** condition is $\int (Q_X - \mu_X) [Q_Y(u) + Q_Y(1-u) - 2\mu_Y] du = 0$.

The **universal (in X)** condition is $Q_Y(u) + Q_Y(1-u)$ constant a.e. (palindromic symmetry in distribution + antisymmetric scoring), exactly mirroring the permutation case.

1.9.6 6) practical diagnostic

Given counts and a fixed score map:

- Expand to sequences (or evaluate on a fine quantile grid for FH),
- Form x' , y' , and $z = y' + \text{rev}(y')$,
- Check $\sum x'z$.
 - If ≈ 0 expect $|r_{\min}| \approx r_{\max}$.
 - If < 0 often $|r_{\min}| > r_{\max}$ (your BES pattern).
 - If > 0 often $|r_{\min}| < r_{\max}$.

1.9.7 tl;dr

- **Pair-specific necessity & sufficiency:** $\sum_i x'_i (y'_i + y'_{n+1-i}) = 0$.
- **Universal (in X) necessity & sufficiency:** one marginal (say Y) has **palindromic mass symmetry** and **antisymmetric scores**—then $|r_{\min}| = r_{\max}$ for all X .

1.9.8 Examples

Below, “scores $0:(K-1)$ ” means category values $[0, 1, \dots, K-1]$. “Affine recode” means $a + b \times (\text{old score})$ with $b > 0$. “Non-affine monotone” means strictly increasing but not affine (e.g., convex spacing).

1.9.9 Notation (permutation extremes, “Lane A”)

- Build expanded vectors x and y from **counts** and **scores**.
- $r_{\max} = \text{cor}(\text{sort}(x, \downarrow), \text{sort}(y, \downarrow))$.
- $r_{\min} = \text{cor}(\text{sort}(x, \downarrow), \text{sort}(y, \uparrow))$.

1.9.10 A) Same number of categories, but $|r_{\min}| \neq r_{\max}$ (concrete numbers)

Counts (both

$$K = 5$$

):

- $X : [100, 200, 300, 200, 200]$
- $Y : [100, 100, 100, 300, 400]$ (skewed; not palindromic)

Scores: 0:4 for both.

- $r_{\max} \approx 0.929398758$
- $r_{\min} \approx -0.881118303$
- $|r_{\min}| \approx 0.8811 \neq r_{\max}$

Affine recode (10,20,30,40,50) for both:

- $r_{\max}, r_{\min}$ **unchanged** (up to rounding).

Non-affine monotone recode for Y : $[0, 1, 1.5, 3, 10]$.

-

$$r_{\max} \approx 0.930261031$$

-

$$r_{\min} \approx -0.863250703$$

- Values change (Pearson is **not** invariant to non-affine spacing); magnitudes still unequal.

Why unequal magnitudes? No palindromic symmetry in Y — the “palindrome-sum” pattern in y' -space isn’t zero, so $S_{\max} + S_{\min} \neq 0$.



1.9.11 B) Different numbers of categories, yet $|r_{\min}| = r_{\max}$ (concrete numbers)

Counts:

- X (4 categories): [100, 200, 300, 400]
- Y (5 categories): [100, 200, 400, 200, 100] (**palindromic**)

Scores: X : 0:3, Y : 0:4.

-

$$r_{\max} \approx 0.912870929$$

-

$$r_{\min} \approx -0.912870929$$

- $|r_{\min}| = r_{\max}$ (to machine precision)

Affine recode (e.g., 10,20,...):

- Magnitudes remain **identical**; affine transforms don’t change Pearson correlation.

Non-affine monotone recode for Y : [0, 1, 1.5, 3, 10].

-

$$r_{\max} \approx 0.580072921$$

-

$$r_{\min} \approx -0.842041337$$

- **Magnitudes no longer match** because antisymmetry of the **scores** about the center is lost, even though the **counts** are palindromic.

Why equal magnitudes in the affine case? Palindromic **mass** symmetry in Y + **anti-symmetric scoring** (any affine map of $0:(K_Y-1)$) the palindrome-sum vector $y'_i + y'_{n+1-i} \equiv 0$. Hence $S_{\max} + S_{\min} = 0$ $|r_{\min}| = r_{\max}$ for **any** X , irrespective of $K_X \neq K_Y$.

1.9.12 C) Spearman invariance (same data as in B)

Compute ranks (with average ties) **once** for x and y , then form extremes by sorting those rank vectors.

- With Y scored 0:4:
 $r_{\max}^{\text{Spearman}} \approx 0.920837215$, $r_{\min}^{\text{Spearman}} \approx -0.920837215$.
- With Y scored non-affinely $[0, 1, 1.5, 3, 10]$:
identical Spearman extremes: 0.920837215 and -0.920837215 .

Spearman is invariant to any strictly monotone recode because ranks depend only on order, not spacing.

1.10 what changes and what doesn't (Pearson vs Spearman)

- **Pearson:**
 - **Affine** recodes $a + b \cdot \text{score}$ (with $b > 0$) leave $r_{\max}, r_{\min}$, and any $|r_{\min}| = r_{\max}$ conclusions **unchanged**.
 - **Non-affine monotone** recodes can change $r_{\max}, r_{\min}$ and can break $|r_{\min}| = r_{\max}$ even with palindromic counts.
 - **Spearman:**
 - Any strictly monotone recode leaves extremes **unchanged**.
 - Palindromic counts in Y (with the usual symmetric rank scheme) give $|r_{\min}| = r_{\max}$.
-

1.11 6. Simulation / Randomization

Permutation distribution:

Generate full vectors from marginals, independently permute X and Y , compute r , repeat.
Produces null distribution centered at 0.

(Proofs-OrdinalVariables.md)

1.12 7. Key Tools Referenced

- **Fréchet–Hoeffding bounds** for extremal joint CDFs.
(BFbounds.md)
 - **Hardy–Littlewood–Pólya rearrangement inequality** for rank-based maximization/minimization.
(Ordinal Variables and Correlation.md)
 - **Comonotone/countermonotone quantile coupling**.
(Math Proof GPT – Whitt 1976 Paper.md)
 - **Capacity limits** in discrete joint tables explain why FH lower may be unattainable.
-

1.13 8. Master Takeaways

1. $r_{\max} = \mathbf{FH\ upper\ (comonotonic)}$, always **attainable** in discrete ordinal settings.
 2. $r_{\min}$ may **not** equal FH lower in discrete settings; use best feasible opposite-order arrangement.
 3. No universal bound on relative magnitudes: $|r_{\min}|$ may be less than, equal to, or greater than $r_{\max}$.
 4. Palindromic symmetry of one marginal guarantees $|r_{\min}| = r_{\max}|$.
 5. Binary cases illustrate when perfect alignment is feasible and when skew prevents it.
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2 Positive Semi-Definiteness in Covariance Matrices: Implications for PCA, Factor Analysis, and SEM

2.1 1. Variance–Covariance Matrices: Definition and Core Property

2.1.0.1 1.1 Definition

For a random vector $\mathbf{X} = (X_1, X_2, \dots, X_n)^\top$, the variance–covariance matrix is

$$= \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top],$$

where $\sigma_{ij} = \text{Cov}(X_i, X_j)$.

(vcov_PSD.md):contentReferenceoaicite:2

- **Diagonal elements** σ_{ii} are variances, always ≥ 0 .
 - **Off-diagonal** entries are covariances.
-

2.1.0.2 1.2 PSD Property

A matrix is **positive semi-definite (PSD)** if, for all nonzero $\mathbf{z}$,

$$\mathbf{z}^\top \mathbf{z} \geq 0.$$

For covariance matrices:

$$\mathbf{z}^\top \mathbf{z} = \mathbb{E}[(\mathbf{z}^\top (\mathbf{X} - \mathbb{E}[\mathbf{X}]))^2] \geq 0,$$

since it is an expectation of a square.

(vcov_PSD.md):contentReferenceoaicite:3

2.1.0.3 1.3 Positive Definite (PD) vs PSD

- **PD** if $\mathbf{z}^\top \mathbf{z} > 0$ for all $\mathbf{z} \neq 0$, i.e., no perfect linear dependence among variables.
 - **PSD** if at least one nonzero $\mathbf{z}$ yields equality (rank-deficient case). *(vcov_PSD.md):contentReferenceoaicite:3*
-

2.2 2. PCA and PSD Requirements

2.2.0.1 2.1 PCA Overview

Principal Components Analysis (PCA) decomposes a sample covariance or correlation matrix $\mathbf{S} = n^{-1}\mathbf{X}^\top\mathbf{X}$ into

$$\mathbf{S} = \mathbf{Q} \text{diag}(\lambda_1, \dots, \lambda_p) \mathbf{Q}^\top.$$

(PCA and FA PSD requirement.md):contentReferenceoaicite:5

2.2.0.2 2.2 Why PSD Matters for PCA

- **Mathematically:** Eigen-decomposition works for any symmetric $\mathbf{S}$, PSD or not.
 - **Interpretively:** Negative eigenvalues break the “variance explained” story (variance cannot be negative).
 - **Software behavior:** Most PCA routines check PD/PSD; failure can cause warnings or dropped components. *(PCA and FA PSD requirement.md):contentReferenceoaicite:6*
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2.2.0.3 2.3 If Input is Symmetric but not PSD

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad \lambda = (3, -1),$$

is indefinite (one negative eigenvalue).

- PCA math still runs; eigenvalues/vectors computed. - Interpretation fails: $\sqrt{\lambda_j}$ imaginary for $\lambda_j < 0$. - Some software refuses input; others run but return nonsensical variance metrics. *(PCA and FA PSD requirement.md):contentReferenceoaicite:7*

2.2.0.4 2.4 Remedies for PCA

1. Treat tiny negative eigenvalues as zero.
 2. Adjust to nearest PSD matrix (Higham 2002).
 3. Run SVD on the raw data matrix instead of precomputed $\mathbf{S}$.
(PCA and FA PSD requirement.md):contentReferenceoaicite:8
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2.3 3. Factor Analysis and PSD Requirements

2.3.0.1 3.1 FA Model

Common factor analysis models Σ as:

$$\Sigma = \Lambda\Lambda^\top + \Psi,$$

where Ψ is diagonal with positive entries. (PCA and FA PSD requirement.md):contentReferenceoaicite:9

2.3.0.2 3.2 PSD/PD Needs in FA

- **ML FA:** Requires $\mathbf{S} \succ 0$; $\log \det \mathbf{S}$ undefined if indefinite.
 - **Principal-axis / Image FA:** Also break with negative eigenvalues (square roots fail).
(PCA and FA PSD requirement.md):contentReferenceoaicite:10
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2.3.0.3 3.3 Remedies for FA

- Repair $\mathbf{S}$ to PSD before analysis.
 - Add small ridge term $\epsilon \mathbf{I}$. (PCA and FA PSD requirement.md):contentReferenceoaicite:11
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2.4 4. SEM and PSD Requirements

2.4.0.1 4.1 SEM Machinery

Observed covariance $\mathbf{S}$; implied covariance $\Sigma(\theta)$ from structural and measurement models. *(PCA and FA PSD requirement.md):contentReferenceoaicite:12*

2.4.0.2 4.2 Fit Functions and PD Needs

- **ML / GLS**: Require both $\mathbf{S}$ and $\Sigma(\theta)$ PD.
 - **WLS/DWLS**: Need PD indirectly (weight matrix invertible).
 - **ULS**: Can start from indefinite $\mathbf{S}$, but test statistics again require PD. *(PCA and FA PSD requirement.md):contentReferenceoaicite:13*
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2.4.0.3 4.3 What Happens if Not PSD

- Software errors out (“matrix is not positive definite”).
 - Log-determinant undefined; inversion fails.
 - Even if bypassed, optimisation steps hit complex/NaN values. *(PCA and FA PSD requirement.md):contentReferenceoaicite:14*
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2.4.0.4 4.4 Remedies in SEM

- Nearest PD smoothing (Higham 2002).
 - Ridge adjustment.
 - Analyse raw data.
 - Check rounding issues. *(PCA and FA PSD requirement.md):contentReferenceoaicite:15*
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2.5 5. Core Lessons Across PCA, FA, SEM

1. **Theoretical PSD:** Covariance matrices from real data are PSD by definition:contentReferenceoaicite:16.
 2. **Numerical Issues:** Sampling noise, rounding, or listwise deletion can yield small negative eigenvalues:contentReferenceoaicite:17.
 3. **PCA:** Can compute from indefinite matrix but interpretation breaks:contentReferenceoaicite:18.
 4. **FA and SEM:** Require PD input; will not run on indefinite matrices:contentReferenceoaicite:19.
 5. **Repairs:** Adjust to nearest PSD, ridge, or recompute from raw data:contentReferenceoaicite:20.
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